

1	Course Name:
	Database Management System
2	Course Code:
	IEB20603
3	Year and Semester Offered:
	Year 3 Semester 1
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis:
	This course introduces students to database management systems theory and design. Practical methodologies for data analysis, data modelling and database design are examined, coupled with a detailed study of the relational database model. Students will build practical skills in data modelling, normalisation and database design. This course presents these topics within a framework that focuses on developing communication and business problem-solving skills.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Demonstrate database design using data models. 2. Apply structured query language (SQL) and advanced SQL. 3. Develop a database system.
8	Content Outline:
	Topic
	1. Introduction to Database
	2. Data Models
	3. The Relational Database Model
	4. Entity Relationship (ER) Model and Advanced Modeling
	5. Normalization of Database Tables
	6. Introduction To Structured Query Language (SQL
	7. Advanced SQL
	8. Database Connectivity and Web Technologies
	9. Transactional Management and Concurrency Control
	10. Big Data and current database trends
	LAB

11. Lab 1: Introduction to database design tools
12. Lab 2: ERD
13. Lab 3: Enhanced ERD
14. Lab 4: Normalisation
15. Lab 5: Introduction to SQL - DML commands
16. Lab 6: SQL - SELECT statements
17. Lab 7: Advanced SQL